RESISTANCE PROJECTION WELDING

Ceramic centering pins + lower electrodes

Selection catalogue for ceramic centering pins and matching threaded lower electrodes for resistance projection welding of weld nuts.

IT-CP
Ceramic centering pins
Short · Long · With collar

IT-ELB
Threaded lower electrodes
Uncooled · Water-cooled



SIZE RANGE	SIZE GROUPS	DISTINCT PART NUMBERS
M4–M12	6	721

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04 Contents
05 M4
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11 M8
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15 M12

RANGE OVERVIEW

The range.

Ceramic centering pins and matching threaded lower electrodes, presented together for accurate centring of weld nuts in resistance projection welding.

721 DISTINCT PART NUMBERS
IT-CP + IT-ELB

PIN · SHORT	PIN · LONG	PIN · WITH COLLAR	ELB · UNCOOLED	ELB · WATER-COOLED	SIZE GROUPS
121	121	121	218	140	6

SIZE	PIN ØD	ELECTRODE ØD	ELB UNCOOLED · G / SW	ELB WATER-COOLED · G / SW
M4	4.5 – 6 mm	4.7 – 6.2 mm	M14x1.5 / 19	M20x1.5 / 30
M5	5.5 – 7 mm	5.7 – 7.2 mm	M14x1.5 / 19 M14x1.5 / 24	M20x1.5 / 30
M6	6.5 – 8.5 mm	6.7 – 8.7 mm	M16x1.5 / 19 M16x1.5 / 24	M20x1.5 / 30
M8	8.5 – 11 mm	8.7 – 11.2 mm	M18x1.5 / 24 M18x1.5 / 30	M20x1.5 / 30
M10	11 – 13 mm	11.2 – 13.2 mm	M20x1.5 / 30	M26x1.5 / 36
M12	13 – 15 mm	13.2 – 15.2 mm	M22x1.5 / 30	M26x1.5 / 36

01 · COORDINATED PIN/ELECTRODE SELECTION

Select the ceramic centering pin and its matching lower electrode from the same row, starting from the hole diameter in the sheet.

02 · SELECTION RULE

Sheet hole Ø − 0.1 mm = pin ØD
Pin ØD + 0.2 mm = electrode ØD

03 · ELB VARIANTS

Uncooled electrodes with different widths across flats (SW) are listed separately, even where ØD is the same.

DGSI / INDUSTRYTOOLS PART NUMBERS

IT-CP / IT-ELB.

Every product carries a primary DGSI / IndustryTools part number. The cross-reference number is provided solely for cross-referencing and compatibility checking.

PRODUCT FAMILY

IT-CP-xxxxx

Ceramic centering pin
Variants: Short · Long · With collar

Cross-reference no. 8228ZKxxxxx

PRODUCT FAMILY

IT-ELB-xxxxx

Threaded lower electrode
Variants: Uncooled · Water-cooled

Cross-reference no. 1111ZKxxxxx

IT-ELS-xxxxx · upper electrodes

RESERVED · NOT INCLUDED IN THIS EDITION

HOW PART NUMBERS ARE SHOWN

IT-ELB-10002
Cross-reference no.: 1111ZK10002

Primary part no. · bold, turquoise
Cross-reference · secondary, grey

SELECTION RULE · M4 EXAMPLE

HOLE Ø IN SHEET	PIN ØD · -0.1 mm	ELECTRODE ØD · +0.2 mm
4.6 mm	→ 4.5 mm	→ 4.7 mm
	IT-CP-11000 · 10000 · 10300	IT-ELB-10002 · 20002

DIMENSIONS USED

ØD	Pin diameter / electrode centre bore	**G**	Electrode external thread
ØD1	Pin body / base diameter	**SW**	Width across flats (SW)
L	Pin length		

Cross-reference numbers are provided for comparison purposes only, to support identification and compatibility checking. They do not indicate that the products are original parts of the respective manufacturer, nor do they imply any affiliation with, approval by or certification from that manufacturer. The primary commercial part number is the DGSI / IndustryTools part number. All trademarks and trade names belong to their respective owners.



CATALOGUE STRUCTURE

Contents.

NO.	SIZE · PINS + ELECTRODES	PIN ØD	ELECTRODE ØD	ROWS	PAGE
01	**M4**	4.5 – 6 mm	4.7 – 6.2 mm	16	05-06
02	**M5**	5.5 – 7 mm	5.7 – 7.2 mm	16	07-08
03	**M6**	6.5 – 8.5 mm	6.7 – 8.7 mm	21	09-10
04	**M8**	8.5 – 11 mm	8.7 – 11.2 mm	26	11-12
05	**M10**	11 – 13 mm	11.2 – 13.2 mm	21	13-14
06	**M12**	13 – 15 mm	13.2 – 15.2 mm	21	15-16

P. 02 Range overview

P. 03 Part numbering IT-CP / IT-ELB

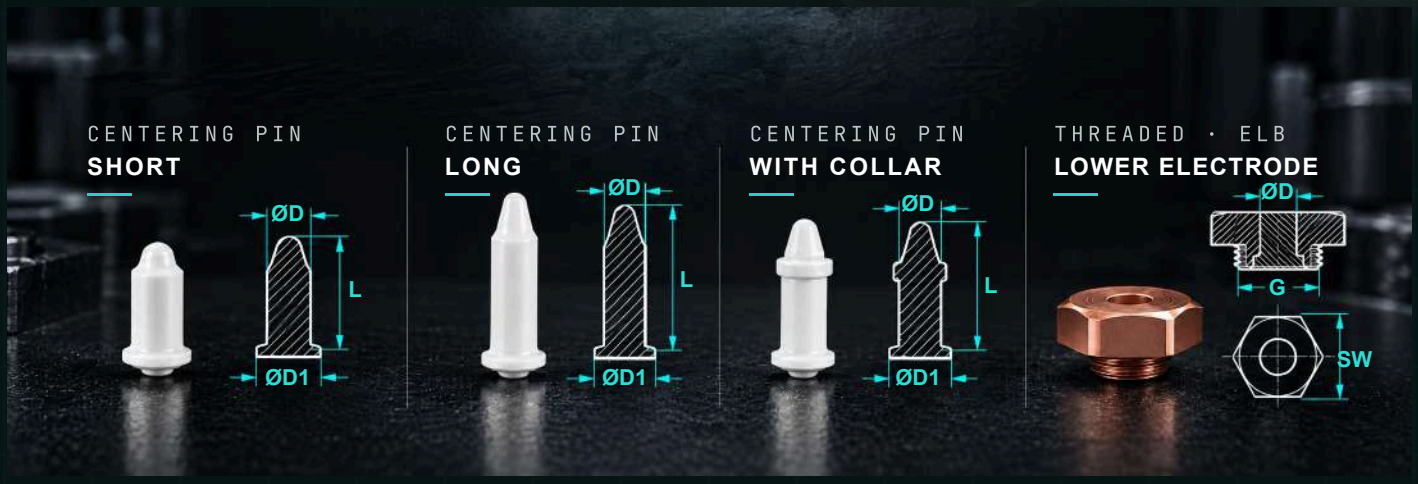
Each size group lists, on a single row, the SHORT, LONG and WITH COLLAR ceramic centering pins together with the matching uncooled and water-cooled threaded lower electrodes, sorted by hole diameter in the sheet. The DGSI / IndustryTools part number is primary; the cross-reference number is shown as secondary information.

GROUP 01 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M4.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	4.6 – 6.1 mm
PIN ØD	4.5 – 6 mm
ELECTRODE ØD	4.7 – 6.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 10 mm	26.5 mm	31 mm / 31 mm	Uncooled M14x1.5 / 19 Water-cooled M20x1.5 / 30

IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.

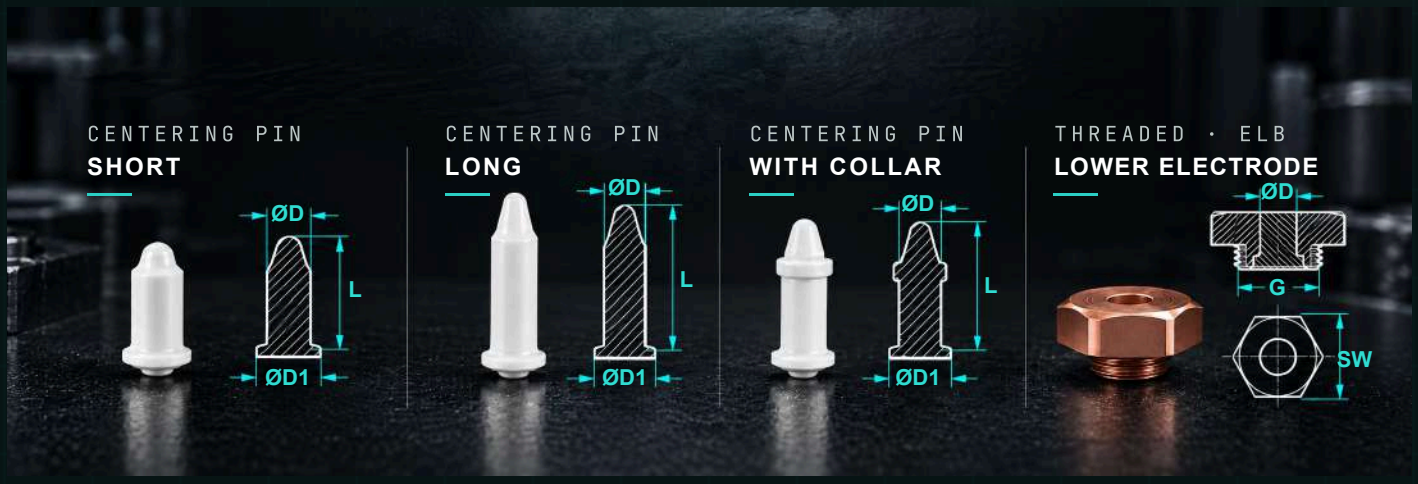
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 26.5 mm	PIN LONG L 31 mm	PIN WITH COLLAR L 31 mm	ELEC-TRODE ØD	ELB UNCOOLED G M14x1.5 · SW 19	ELB WATER-COOLED G M20x1.5 · SW 30
4.6	4.5	IT-CP-11000 8228ZK11000	IT-CP-10000 8228ZK10000	IT-CP-10300 8228ZK10300	4.7	IT-ELB-10002 1111ZK10002	IT-ELB-20002 1111ZK20002
4.7	4.6	IT-CP-11001 8228ZK11001	IT-CP-10001 8228ZK10001	IT-CP-10301 8228ZK10301	4.8	IT-ELB-10003 1111ZK10003	IT-ELB-20003 1111ZK20003
4.8	4.7	IT-CP-11002 8228ZK11002	IT-CP-10002 8228ZK10002	IT-CP-10302 8228ZK10302	4.9	IT-ELB-10004 1111ZK10004	IT-ELB-20004 1111ZK20004
4.9	4.8	IT-CP-11003 8228ZK11003	IT-CP-10003 8228ZK10003	IT-CP-10303 8228ZK10303	5	IT-ELB-10005 1111ZK10005	IT-ELB-20005 1111ZK20005
5	4.9	IT-CP-11004 8228ZK11004	IT-CP-10004 8228ZK10004	IT-CP-10304 8228ZK10304	5.1	IT-ELB-10006 1111ZK10006	IT-ELB-20006 1111ZK20006
5.1	5	IT-CP-11005 8228ZK11005	IT-CP-10005 8228ZK10005	IT-CP-10305 8228ZK10305	5.2	IT-ELB-10007 1111ZK10007	IT-ELB-20007 1111ZK20007
5.2	5.1	IT-CP-11006 8228ZK11006	IT-CP-10006 8228ZK10006	IT-CP-10306 8228ZK10306	5.3	IT-ELB-10008 1111ZK10008	IT-ELB-20008 1111ZK20008
5.3	5.2	IT-CP-11007 8228ZK11007	IT-CP-10007 8228ZK10007	IT-CP-10307 8228ZK10307	5.4	IT-ELB-10009 1111ZK10009	IT-ELB-20009 1111ZK20009

GROUP 01 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M4.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	4.6 – 6.1 mm
PIN ØD	4.5 – 6 mm
ELECTRODE ØD	4.7 – 6.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 10 mm	26.5 mm	31 mm / 31 mm	Uncooled M14x1.5 / 19 Water-cooled M20x1.5 / 30

IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.

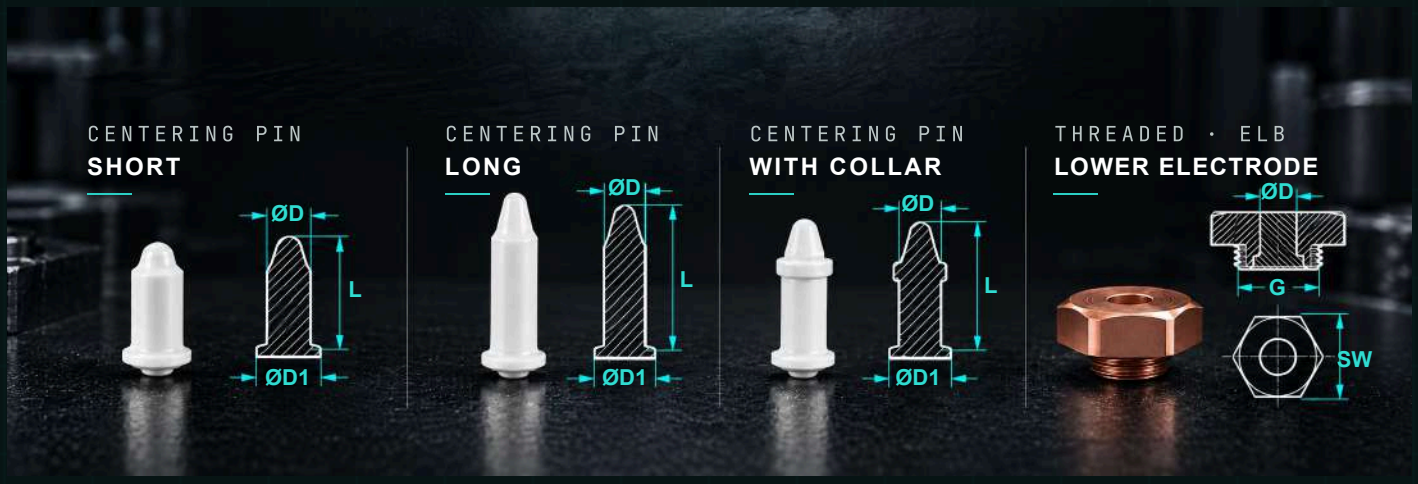
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 26.5 mm	PIN LONG L 31 mm	PIN WITH COLLAR L 31 mm	ELEC-TRODE ØD	ELB UNCOOLED G M14x1.5 · SW 19	ELB WATER-COOLED G M20x1.5 · SW 30
5.4	5.3	IT-CP-11008 8228ZK11008	IT-CP-10008 8228ZK10008	IT-CP-10308 8228ZK10308	5.5	IT-ELB-10010 1111ZK10010	IT-ELB-20010 1111ZK20010
5.5	5.4	IT-CP-11009 8228ZK11009	IT-CP-10009 8228ZK10009	IT-CP-10309 8228ZK10309	5.6	IT-ELB-10011 1111ZK10011	IT-ELB-20011 1111ZK20011
5.6	5.5	IT-CP-11010 8228ZK11010	IT-CP-10010 8228ZK10010	IT-CP-10310 8228ZK10310	5.7	IT-ELB-10012 1111ZK10012	IT-ELB-20012 1111ZK20012
5.7	5.6	IT-CP-11011 8228ZK11011	IT-CP-10011 8228ZK10011	IT-CP-10311 8228ZK10311	5.8	IT-ELB-10013 1111ZK10013	IT-ELB-20013 1111ZK20013
5.8	5.7	IT-CP-11012 8228ZK11012	IT-CP-10012 8228ZK10012	IT-CP-10312 8228ZK10312	5.9	IT-ELB-10014 1111ZK10014	IT-ELB-20014 1111ZK20014
5.9	5.8	IT-CP-11013 8228ZK11013	IT-CP-10013 8228ZK10013	IT-CP-10313 8228ZK10313	6	IT-ELB-10015 1111ZK10015	IT-ELB-20015 1111ZK20015
6	5.9	IT-CP-11014 8228ZK11014	IT-CP-10014 8228ZK10014	IT-CP-10314 8228ZK10314	6.1	IT-ELB-10016 1111ZK10016	IT-ELB-20016 1111ZK20016
6.1	6	IT-CP-11015 8228ZK11015	IT-CP-10015 8228ZK10015	IT-CP-10315 8228ZK10315	6.2	IT-ELB-10017 1111ZK10017	IT-ELB-20017 1111ZK20017

GROUP 02 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M5.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	5.6 – 7.1 mm
PIN ØD	5.5 – 7 mm
ELECTRODE ØD	5.7 – 7.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS
ØD1 10 mm

L · SHORT
26.5 mm

L · LONG / WITH COLLAR
31 mm / 31 mm

ELB · G / SW
Uncooled M14x1.5 / 19 · M14x1.5 / 24
Water-cooled M20x1.5 / 30

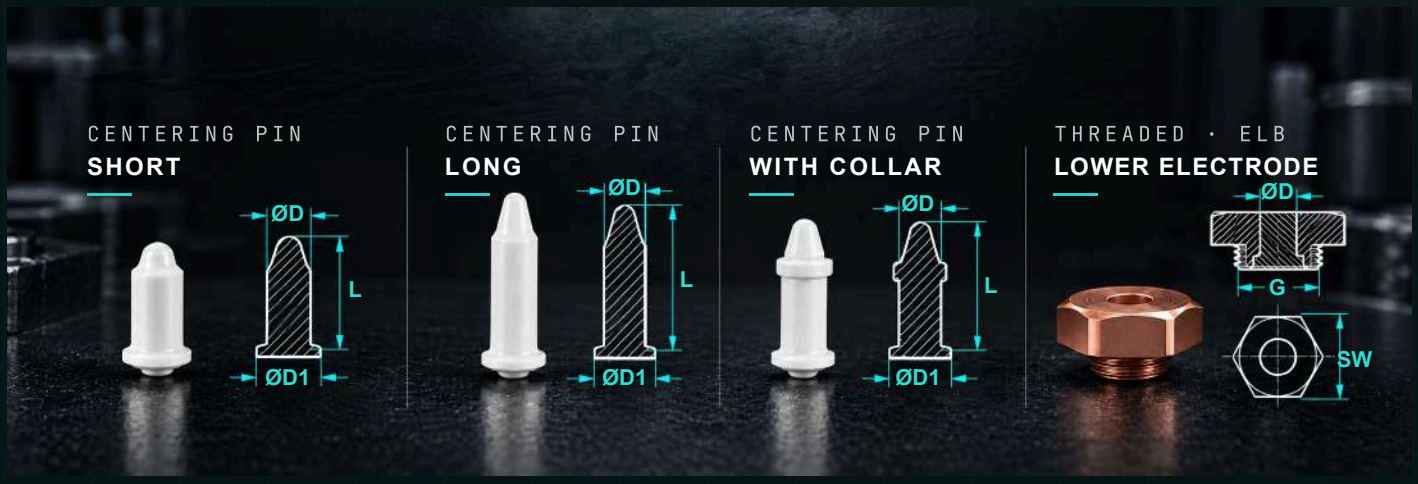
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 26.5 mm	PIN LONG L 31 mm	PIN WITH COLLAR L 31 mm	ELEC-TRODE ØD	ELB UNCOOLED G M14x1.5 · SW 19	ELB UNCOOLED G M14x1.5 · SW 24	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.						
5.6	5.5	IT-CP-11016 8228ZK11016	IT-CP-10016 8228ZK10016	IT-CP-10316 8228ZK10316	5.7	IT-ELB-10012 1111ZK10012	IT-ELB-10059 1111ZK10059	IT-ELB-20012 1111ZK20012
5.7	5.6	IT-CP-11017 8228ZK11017	IT-CP-10017 8228ZK10017	IT-CP-10317 8228ZK10317	5.8	IT-ELB-10013 1111ZK10013	IT-ELB-10060 1111ZK10060	IT-ELB-20013 1111ZK20013
5.8	5.7	IT-CP-11018 8228ZK11018	IT-CP-10018 8228ZK10018	IT-CP-10318 8228ZK10318	5.9	IT-ELB-10014 1111ZK10014	IT-ELB-10061 1111ZK10061	IT-ELB-20014 1111ZK20014
5.9	5.8	IT-CP-11019 8228ZK11019	IT-CP-10019 8228ZK10019	IT-CP-10319 8228ZK10319	6	IT-ELB-10015 1111ZK10015	IT-ELB-10062 1111ZK10062	IT-ELB-20015 1111ZK20015
6	5.9	IT-CP-11020 8228ZK11020	IT-CP-10020 8228ZK10020	IT-CP-10320 8228ZK10320	6.1	IT-ELB-10016 1111ZK10016	IT-ELB-10063 1111ZK10063	IT-ELB-20016 1111ZK20016
6.1	6	IT-CP-11021 8228ZK11021	IT-CP-10021 8228ZK10021	IT-CP-10321 8228ZK10321	6.2	IT-ELB-10017 1111ZK10017	IT-ELB-10064 1111ZK10064	IT-ELB-20017 1111ZK20017
6.2	6.1	IT-CP-11022 8228ZK11022	IT-CP-10022 8228ZK10022	IT-CP-10322 8228ZK10322	6.3	IT-ELB-10018 1111ZK10018	IT-ELB-10065 1111ZK10065	IT-ELB-20018 1111ZK20018
6.3	6.2	IT-CP-11023 8228ZK11023	IT-CP-10023 8228ZK10023	IT-CP-10323 8228ZK10323	6.4	IT-ELB-10019 1111ZK10019	IT-ELB-10066 1111ZK10066	IT-ELB-20019 1111ZK20019

GROUP 02 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M5.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	5.6 – 7.1 mm
PIN ØD	5.5 – 7 mm
ELECTRODE ØD	5.7 – 7.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 10 mm	26.5 mm	31 mm / 31 mm	Uncooled M14x1.5 / 19 · M14x1.5 / 24 Water-cooled M20x1.5 / 30

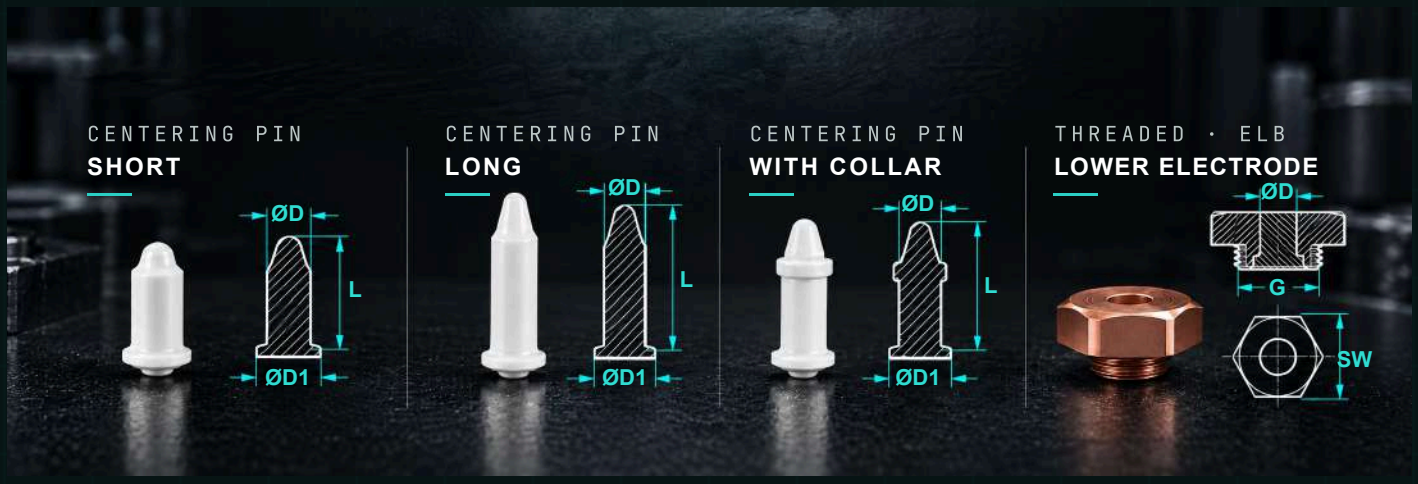
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 26.5 mm	PIN LONG L 31 mm	PIN WITH COLLAR L 31 mm	ELEC-TRODE ØD	ELB UNCOOLED G M14x1.5 · SW 19	ELB UNCOOLED G M14x1.5 · SW 24	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no.				8228… / 1111… Cross-reference no.		
6.4	6.3	IT-CP-11024 8228ZK11024	IT-CP-10024 8228ZK10024	IT-CP-10324 8228ZK10324	6.5	IT-ELB-10020 1111ZK10020	IT-ELB-10067 1111ZK10067	IT-ELB-20020 1111ZK20020
6.5	6.4	IT-CP-11025 8228ZK11025	IT-CP-10025 8228ZK10025	IT-CP-10325 8228ZK10325	6.6	IT-ELB-10021 1111ZK10021	IT-ELB-10068 1111ZK10068	IT-ELB-20021 1111ZK20021
6.6	6.5	IT-CP-11026 8228ZK11026	IT-CP-10026 8228ZK10026	IT-CP-10326 8228ZK10326	6.7	IT-ELB-10022 1111ZK10022	IT-ELB-10069 1111ZK10069	IT-ELB-20022 1111ZK20022
6.7	6.6	IT-CP-11027 8228ZK11027	IT-CP-10027 8228ZK10027	IT-CP-10327 8228ZK10327	6.8	IT-ELB-10023 1111ZK10023	IT-ELB-10070 1111ZK10070	IT-ELB-20023 1111ZK20023
6.8	6.7	IT-CP-11028 8228ZK11028	IT-CP-10028 8228ZK10028	IT-CP-10328 8228ZK10328	6.9	IT-ELB-10024 1111ZK10024	IT-ELB-10071 1111ZK10071	IT-ELB-20024 1111ZK20024
6.9	6.8	IT-CP-11029 8228ZK11029	IT-CP-10029 8228ZK10029	IT-CP-10329 8228ZK10329	7	IT-ELB-10025 1111ZK10025	IT-ELB-10072 1111ZK10072	IT-ELB-20025 1111ZK20025
7	6.9	IT-CP-11030 8228ZK11030	IT-CP-10030 8228ZK10030	IT-CP-10330 8228ZK10330	7.1	IT-ELB-10026 1111ZK10026	IT-ELB-10073 1111ZK10073	IT-ELB-20026 1111ZK20026
7.1	7	IT-CP-11031 8228ZK11031	IT-CP-10031 8228ZK10031	IT-CP-10331 8228ZK10331	7.2	IT-ELB-10027 1111ZK10027	IT-ELB-10074 1111ZK10074	IT-ELB-20027 1111ZK20027

GROUP 03 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M6.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	6.6 – 8.6 mm
PIN ØD	6.5 – 8.5 mm
ELECTRODE ØD	6.7 – 8.7 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 11.5 mm	27.5 mm	34 mm / 34 mm	Uncooled M16x1.5 / 19 · M16x1.5 / 24 Water-cooled M20x1.5 / 30

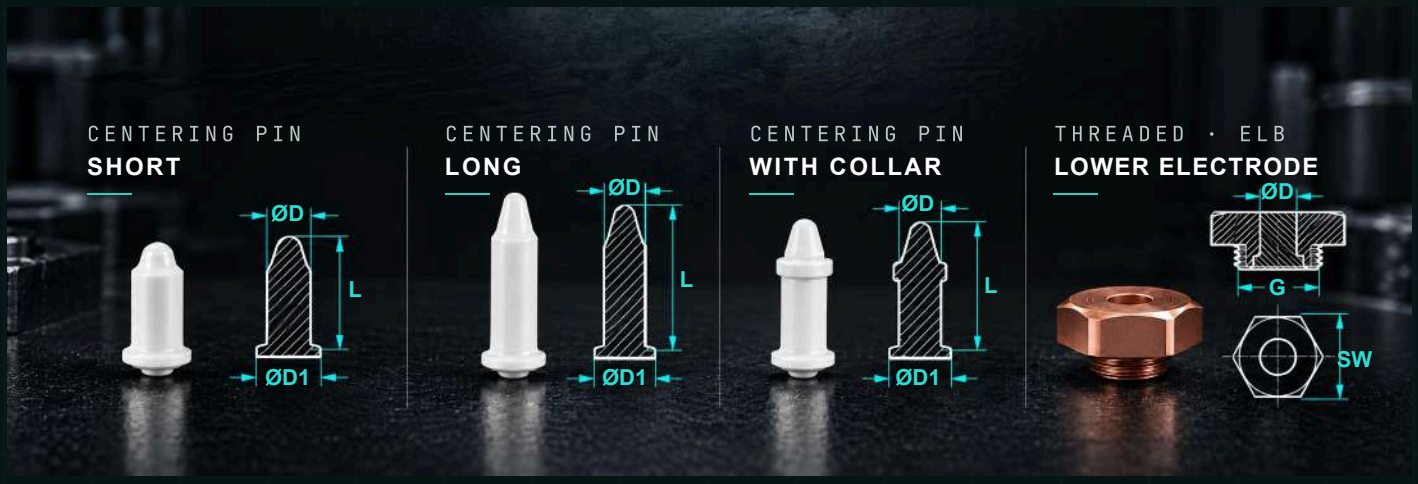
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 27.5 mm	PIN LONG L 34 mm	PIN WITH COLLAR L 34 mm	ELEC-TRODE ØD	ELB UNCOOLED G M16x1.5 · SW 19	ELB UNCOOLED G M16x1.5 · SW 24	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no.				8228… / 1111… Cross-reference no.		
6.6	6.5	IT-CP-11032 8228ZK11032	IT-CP-10032 8228ZK10032	IT-CP-10332 8228ZK10332	6.7	IT-ELB-10033 1111ZK10033	IT-ELB-10080 1111ZK10080	IT-ELB-20033 1111ZK20033
6.7	6.6	IT-CP-11033 8228ZK11033	IT-CP-10033 8228ZK10033	IT-CP-10333 8228ZK10333	6.8	IT-ELB-10034 1111ZK10034	IT-ELB-10081 1111ZK10081	IT-ELB-20034 1111ZK20034
6.8	6.7	IT-CP-11034 8228ZK11034	IT-CP-10034 8228ZK10034	IT-CP-10334 8228ZK10334	6.9	IT-ELB-10035 1111ZK10035	IT-ELB-10082 1111ZK10082	IT-ELB-20035 1111ZK20035
6.9	6.8	IT-CP-11035 8228ZK11035	IT-CP-10035 8228ZK10035	IT-CP-10335 8228ZK10335	7	IT-ELB-10036 1111ZK10036	IT-ELB-10083 1111ZK10083	IT-ELB-20036 1111ZK20036
7	6.9	IT-CP-11036 8228ZK11036	IT-CP-10036 8228ZK10036	IT-CP-10336 8228ZK10336	7.1	IT-ELB-10037 1111ZK10037	IT-ELB-10084 1111ZK10084	IT-ELB-20037 1111ZK20037
7.1	7	IT-CP-11037 8228ZK11037	IT-CP-10037 8228ZK10037	IT-CP-10337 8228ZK10337	7.2	IT-ELB-10038 1111ZK10038	IT-ELB-10085 1111ZK10085	IT-ELB-20038 1111ZK20038
7.2	7.1	IT-CP-11038 8228ZK11038	IT-CP-10038 8228ZK10038	IT-CP-10338 8228ZK10338	7.3	IT-ELB-10039 1111ZK10039	IT-ELB-10086 1111ZK10086	IT-ELB-20039 1111ZK20039
7.3	7.2	IT-CP-11039 8228ZK11039	IT-CP-10039 8228ZK10039	IT-CP-10339 8228ZK10339	7.4	IT-ELB-10040 1111ZK10040	IT-ELB-10087 1111ZK10087	IT-ELB-20040 1111ZK20040
7.4	7.3	IT-CP-11040 8228ZK11040	IT-CP-10040 8228ZK10040	IT-CP-10340 8228ZK10340	7.5	IT-ELB-10041 1111ZK10041	IT-ELB-10088 1111ZK10088	IT-ELB-20041 1111ZK20041
7.5	7.4	IT-CP-11041 8228ZK11041	IT-CP-10041 8228ZK10041	IT-CP-10341 8228ZK10341	7.6	IT-ELB-10042 1111ZK10042	IT-ELB-10089 1111ZK10089	IT-ELB-20042 1111ZK20042
7.6	7.5	IT-CP-11042 8228ZK11042	IT-CP-10042 8228ZK10042	IT-CP-10342 8228ZK10342	7.7	IT-ELB-10043 1111ZK10043	IT-ELB-10090 1111ZK10090	IT-ELB-20043 1111ZK20043

GROUP 03 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M6.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	6.6 – 8.6 mm
PIN ØD	6.5 – 8.5 mm
ELECTRODE ØD	6.7 – 8.7 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS
ØD1 11.5 mm

L · SHORT
27.5 mm

L · LONG / WITH COLLAR
34 mm / 34 mm

ELB · G / SW
Uncooled M16x1.5 / 19 · M16x1.5 / 24
Water-cooled M20x1.5 / 30

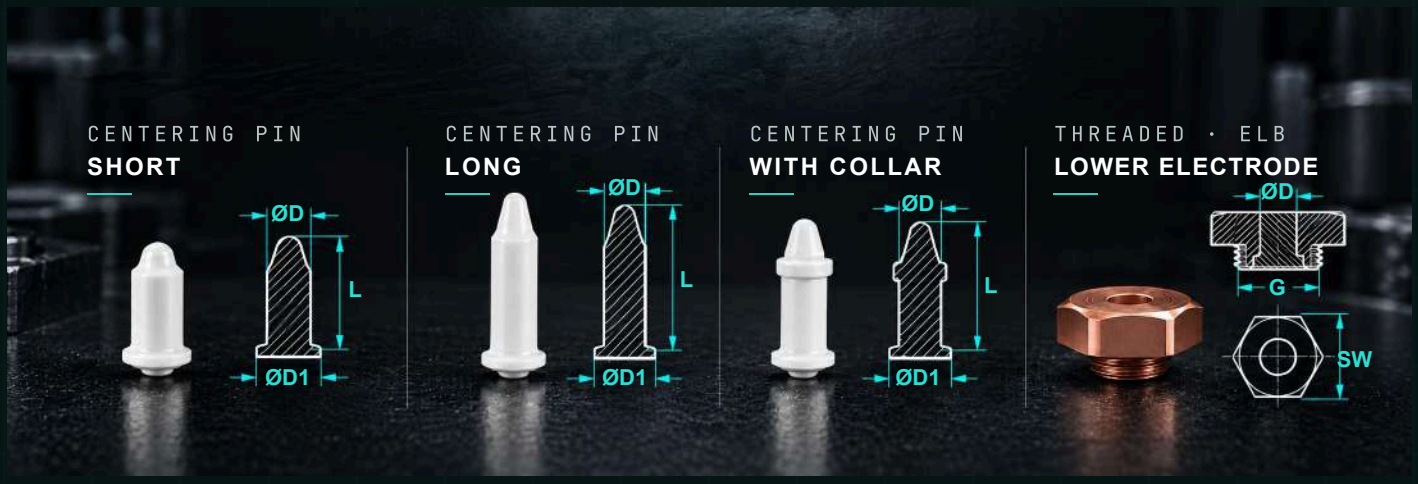
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 27.5 mm	PIN LONG L 34 mm	PIN WITH COLLAR L 34 mm	ELEC-TRODE ØD	ELB UNCOOLED G M16x1.5 · SW 19	ELB UNCOOLED G M16x1.5 · SW 24	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.						
7.7	7.6	IT-CP-11043 8228ZK11043	IT-CP-10043 8228ZK10043	IT-CP-10343 8228ZK10343	7.8	IT-ELB-10044 1111ZK10044	IT-ELB-10091 1111ZK10091	IT-ELB-20044 1111ZK20044
7.8	7.7	IT-CP-11044 8228ZK11044	IT-CP-10044 8228ZK10044	IT-CP-10344 8228ZK10344	7.9	IT-ELB-10045 1111ZK10045	IT-ELB-10092 1111ZK10092	IT-ELB-20045 1111ZK20045
7.9	7.8	IT-CP-11045 8228ZK11045	IT-CP-10045 8228ZK10045	IT-CP-10345 8228ZK10345	8	IT-ELB-10046 1111ZK10046	IT-ELB-10093 1111ZK10093	IT-ELB-20046 1111ZK20046
8	7.9	IT-CP-11046 8228ZK11046	IT-CP-10046 8228ZK10046	IT-CP-10346 8228ZK10346	8.1	IT-ELB-10047 1111ZK10047	IT-ELB-10094 1111ZK10094	IT-ELB-20047 1111ZK20047
8.1	8	IT-CP-11047 8228ZK11047	IT-CP-10047 8228ZK10047	IT-CP-10347 8228ZK10347	8.2	IT-ELB-10048 1111ZK10048	IT-ELB-10095 1111ZK10095	IT-ELB-20048 1111ZK20048
8.2	8.1	IT-CP-11048 8228ZK11048	IT-CP-10048 8228ZK10048	IT-CP-10348 8228ZK10348	8.3	IT-ELB-10049 1111ZK10049	IT-ELB-10096 1111ZK10096	IT-ELB-20049 1111ZK20049
8.3	8.2	IT-CP-11049 8228ZK11049	IT-CP-10049 8228ZK10049	IT-CP-10349 8228ZK10349	8.4	IT-ELB-10050 1111ZK10050	IT-ELB-10097 1111ZK10097	IT-ELB-20050 1111ZK20050
8.4	8.3	IT-CP-11050 8228ZK11050	IT-CP-10050 8228ZK10050	IT-CP-10350 8228ZK10350	8.5	IT-ELB-10051 1111ZK10051	IT-ELB-10098 1111ZK10098	IT-ELB-20051 1111ZK20051
8.5	8.4	IT-CP-11051 8228ZK11051	IT-CP-10051 8228ZK10051	IT-CP-10351 8228ZK10351	8.6	IT-ELB-10052 1111ZK10052	IT-ELB-10099 1111ZK10099	IT-ELB-20052 1111ZK20052
8.6	8.5	IT-CP-11052 8228ZK11052	IT-CP-10052 8228ZK10052	IT-CP-10352 8228ZK10352	8.7	IT-ELB-10053 1111ZK10053	IT-ELB-10100 1111ZK10100	IT-ELB-20053 1111ZK20053

GROUP 04 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M8.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	8.6 – 11.1 mm
PIN ØD	8.5 – 11 mm
ELECTRODE ØD	8.7 – 11.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS
ØD1 13.5 mm

L · SHORT
29.8 mm

L · LONG / WITH COLLAR
35 mm / 35 mm

ELB · G / SW
Uncooled M18x1.5 / 24 · M18x1.5 / 30
Water-cooled M20x1.5 / 30

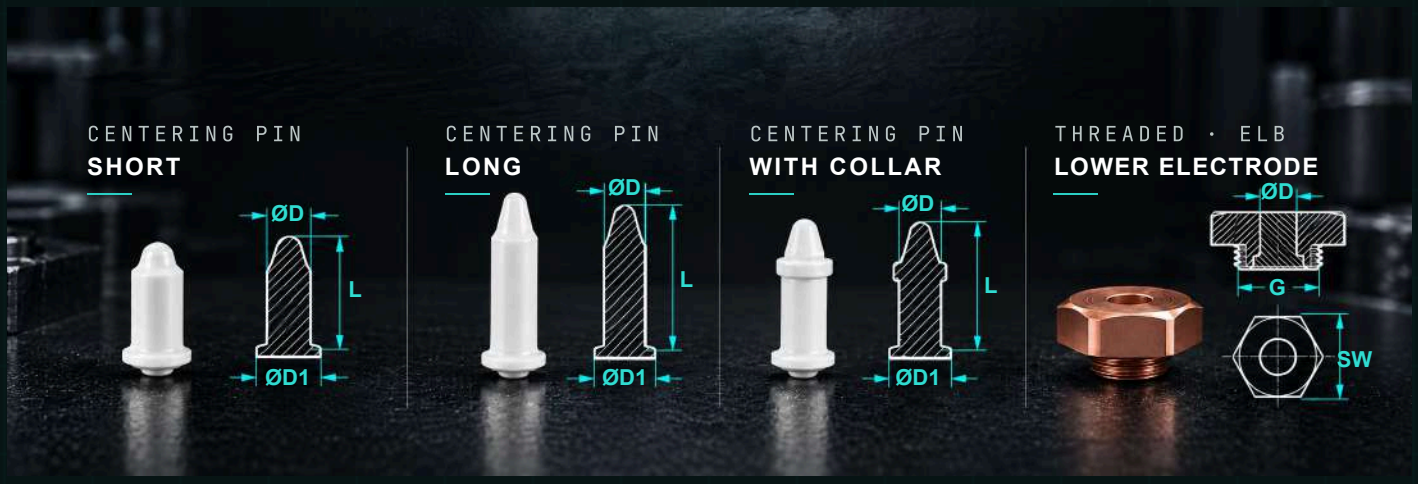
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 29.8 mm	PIN LONG L 35 mm	PIN WITH COLLAR L 35 mm	ELEC-TRODE ØD	ELB UNCOOLED G M18x1.5 · SW 24	ELB UNCOOLED G M18x1.5 · SW 30	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no.				8228… / 1111… Cross-reference no.		
8.6	8.5	IT-CP-11053 8228ZK11053	IT-CP-10053 8228ZK10053	IT-CP-10353 8228ZK10353	8.7	IT-ELB-10106 1111ZK10106	IT-ELB-10137 1111ZK10137	IT-ELB-20106 1111ZK20106
8.7	8.6	IT-CP-11054 8228ZK11054	IT-CP-10054 8228ZK10054	IT-CP-10354 8228ZK10354	8.8	IT-ELB-10107 1111ZK10107	IT-ELB-10138 1111ZK10138	IT-ELB-20107 1111ZK20107
8.8	8.7	IT-CP-11055 8228ZK11055	IT-CP-10055 8228ZK10055	IT-CP-10355 8228ZK10355	8.9	IT-ELB-10108 1111ZK10108	IT-ELB-10139 1111ZK10139	IT-ELB-20108 1111ZK20108
8.9	8.8	IT-CP-11056 8228ZK11056	IT-CP-10056 8228ZK10056	IT-CP-10356 8228ZK10356	9	IT-ELB-10109 1111ZK10109	IT-ELB-10140 1111ZK10140	IT-ELB-20109 1111ZK20109
9	8.9	IT-CP-11057 8228ZK11057	IT-CP-10057 8228ZK10057	IT-CP-10357 8228ZK10357	9.1	IT-ELB-10110 1111ZK10110	IT-ELB-10141 1111ZK10141	IT-ELB-20110 1111ZK20110
9.1	9	IT-CP-11058 8228ZK11058	IT-CP-10058 8228ZK10058	IT-CP-10358 8228ZK10358	9.2	IT-ELB-10111 1111ZK10111	IT-ELB-10142 1111ZK10142	IT-ELB-20111 1111ZK20111
9.2	9.1	IT-CP-11059 8228ZK11059	IT-CP-10059 8228ZK10059	IT-CP-10359 8228ZK10359	9.3	IT-ELB-10112 1111ZK10112	IT-ELB-10143 1111ZK10143	IT-ELB-20112 1111ZK20112
9.3	9.2	IT-CP-11060 8228ZK11060	IT-CP-10060 8228ZK10060	IT-CP-10360 8228ZK10360	9.4	IT-ELB-10113 1111ZK10113	IT-ELB-10144 1111ZK10144	IT-ELB-20113 1111ZK20113
9.4	9.3	IT-CP-11061 8228ZK11061	IT-CP-10061 8228ZK10061	IT-CP-10361 8228ZK10361	9.5	IT-ELB-10114 1111ZK10114	IT-ELB-10145 1111ZK10145	IT-ELB-20114 1111ZK20114
9.5	9.4	IT-CP-11062 8228ZK11062	IT-CP-10062 8228ZK10062	IT-CP-10362 8228ZK10362	9.6	IT-ELB-10115 1111ZK10115	IT-ELB-10146 1111ZK10146	IT-ELB-20115 1111ZK20115
9.6	9.5	IT-CP-11063 8228ZK11063	IT-CP-10063 8228ZK10063	IT-CP-10363 8228ZK10363	9.7	IT-ELB-10116 1111ZK10116	IT-ELB-10147 1111ZK10147	IT-ELB-20116 1111ZK20116
9.7	9.6	IT-CP-11064 8228ZK11064	IT-CP-10064 8228ZK10064	IT-CP-10364 8228ZK10364	9.8	IT-ELB-10117 1111ZK10117	IT-ELB-10148 1111ZK10148	IT-ELB-20117 1111ZK20117
9.8	9.7	IT-CP-11065 8228ZK11065	IT-CP-10065 8228ZK10065	IT-CP-10365 8228ZK10365	9.9	IT-ELB-10118 1111ZK10118	IT-ELB-10149 1111ZK10149	IT-ELB-20118 1111ZK20118

GROUP 04 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M8.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	8.6 – 11.1 mm
PIN ØD	8.5 – 11 mm
ELECTRODE ØD	8.7 – 11.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 13.5 mm	29.8 mm	35 mm / 35 mm	Uncooled M18x1.5 / 24 · M18x1.5 / 30 Water-cooled M20x1.5 / 30

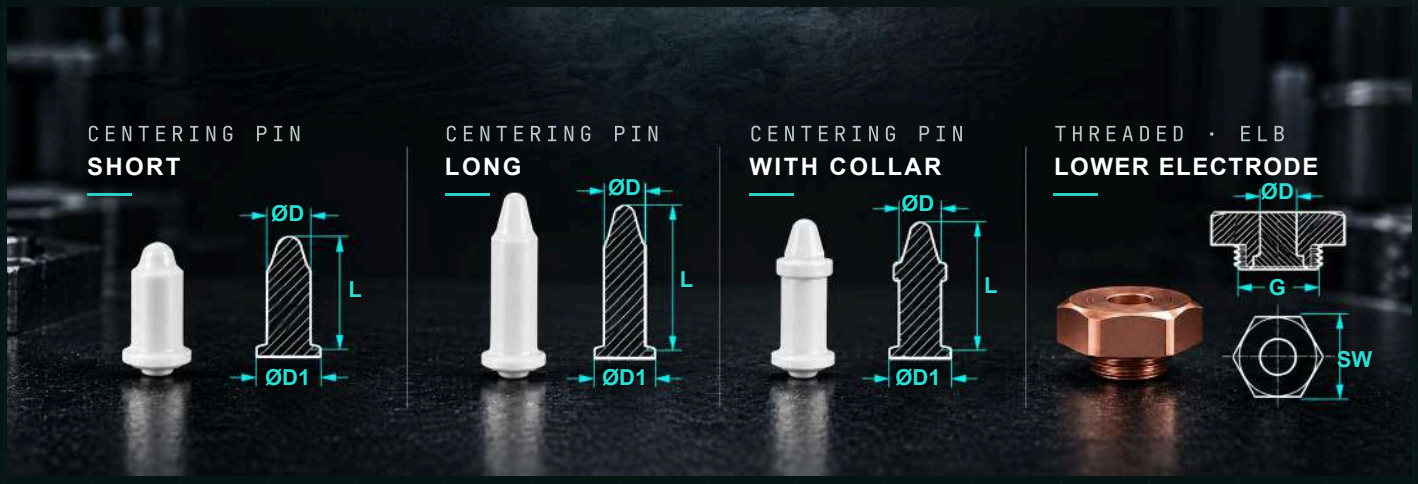
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 29.8 mm	PIN LONG L 35 mm	PIN WITH COLLAR L 35 mm	ELEC-TRODE ØD	ELB UNCOOLED G M18x1.5 · SW 24	ELB UNCOOLED G M18x1.5 · SW 30	ELB WATER-COOLED G M20x1.5 · SW 30
		IT-… DGSI / IndustryTools part no.					8228… / 1111… Cross-reference no.	
9.9	9.8	IT-CP-11066 8228ZK11066	IT-CP-10066 8228ZK10066	IT-CP-10366 8228ZK10366	10	IT-ELB-10119 1111ZK10119	IT-ELB-10150 1111ZK10150	IT-ELB-20119 1111ZK20119
10	9.9	IT-CP-11067 8228ZK11067	IT-CP-10067 8228ZK10067	IT-CP-10367 8228ZK10367	10.1	IT-ELB-10120 1111ZK10120	IT-ELB-10151 1111ZK10151	IT-ELB-20120 1111ZK20120
10.1	10	IT-CP-11068 8228ZK11068	IT-CP-10068 8228ZK10068	IT-CP-10368 8228ZK10368	10.2	IT-ELB-10121 1111ZK10121	IT-ELB-10152 1111ZK10152	IT-ELB-20121 1111ZK20121
10.2	10.1	IT-CP-11069 8228ZK11069	IT-CP-10069 8228ZK10069	IT-CP-10369 8228ZK10369	10.3	IT-ELB-10122 1111ZK10122	IT-ELB-10153 1111ZK10153	IT-ELB-20122 1111ZK20122
10.3	10.2	IT-CP-11070 8228ZK11070	IT-CP-10070 8228ZK10070	IT-CP-10370 8228ZK10370	10.4	IT-ELB-10123 1111ZK10123	IT-ELB-10154 1111ZK10154	IT-ELB-20123 1111ZK20123
10.4	10.3	IT-CP-11071 8228ZK11071	IT-CP-10071 8228ZK10071	IT-CP-10371 8228ZK10371	10.5	IT-ELB-10124 1111ZK10124	IT-ELB-10155 1111ZK10155	IT-ELB-20124 1111ZK20124
10.5	10.4	IT-CP-11072 8228ZK11072	IT-CP-10072 8228ZK10072	IT-CP-10372 8228ZK10372	10.6	IT-ELB-10125 1111ZK10125	IT-ELB-10156 1111ZK10156	IT-ELB-20125 1111ZK20125
10.6	10.5	IT-CP-11073 8228ZK11073	IT-CP-10073 8228ZK10073	IT-CP-10373 8228ZK10373	10.7	IT-ELB-10126 1111ZK10126	IT-ELB-10157 1111ZK10157	IT-ELB-20126 1111ZK20126
10.7	10.6	IT-CP-11074 8228ZK11074	IT-CP-10074 8228ZK10074	IT-CP-10374 8228ZK10374	10.8	IT-ELB-10127 1111ZK10127	IT-ELB-10158 1111ZK10158	IT-ELB-20127 1111ZK20127
10.8	10.7	IT-CP-11075 8228ZK11075	IT-CP-10075 8228ZK10075	IT-CP-10375 8228ZK10375	10.9	IT-ELB-10128 1111ZK10128	IT-ELB-10159 1111ZK10159	IT-ELB-20128 1111ZK20128
10.9	10.8	IT-CP-11076 8228ZK11076	IT-CP-10076 8228ZK10076	IT-CP-10376 8228ZK10376	11	IT-ELB-10129 1111ZK10129	IT-ELB-10160 1111ZK10160	IT-ELB-20129 1111ZK20129
11	10.9	IT-CP-11077 8228ZK11077	IT-CP-10077 8228ZK10077	IT-CP-10377 8228ZK10377	11.1	IT-ELB-10130 1111ZK10130	IT-ELB-10161 1111ZK10161	IT-ELB-20130 1111ZK20130
11.1	11	IT-CP-11078 8228ZK11078	IT-CP-10078 8228ZK10078	IT-CP-10378 8228ZK10378	11.2	IT-ELB-10131 1111ZK10131	IT-ELB-10162 1111ZK10162	IT-ELB-20131 1111ZK20131

GROUP 05 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M10.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	11.1 – 13.1 mm
PIN ØD	11 – 13 mm
ELECTRODE ØD	11.2 – 13.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 16 mm	32.3 mm	40 mm / 40 mm	Uncooled M20x1.5 / 30 Water-cooled M26x1.5 / 36

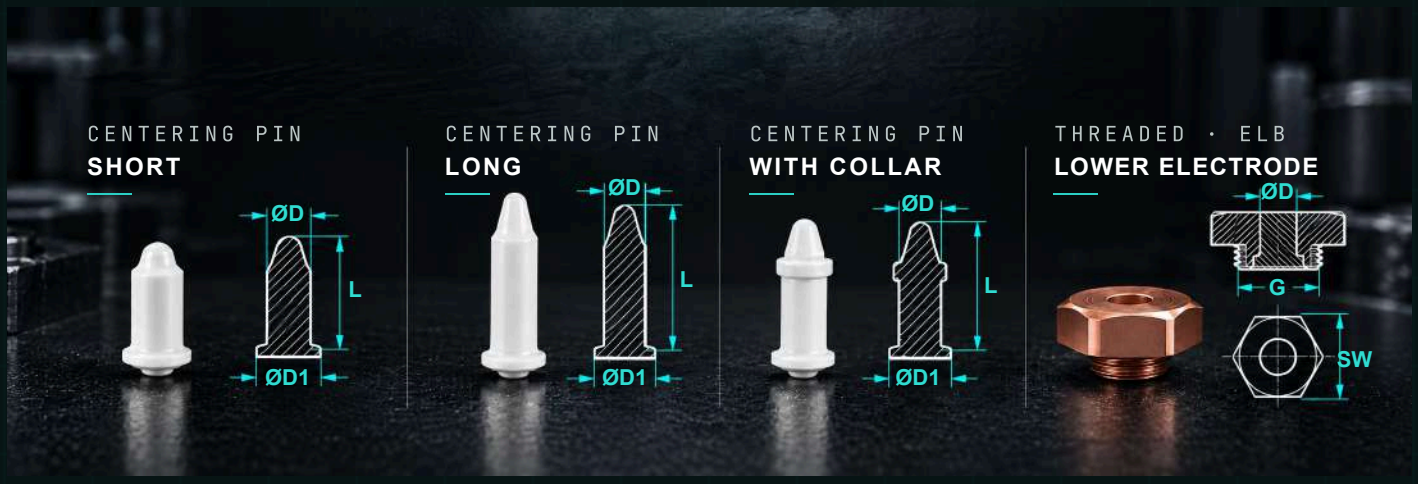
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 32.3 mm	PIN LONG L 40 mm	PIN WITH COLLAR L 40 mm	ELEC-TRODE ØD	ELB UNCOOLED G M20x1.5 · SW 30	ELB WATER-COOLED G M26x1.5 · SW 36
		IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.					
11.1	11	IT-CP-11079 8228ZK11079	IT-CP-10079 8228ZK10079	IT-CP-10379 8228ZK10379	11.2	IT-ELB-10168 1111ZK10168	IT-ELB-20168 1111ZK20168
11.2	11.1	IT-CP-11080 8228ZK11080	IT-CP-10080 8228ZK10080	IT-CP-10380 8228ZK10380	11.3	IT-ELB-10169 1111ZK10169	IT-ELB-20169 1111ZK20169
11.3	11.2	IT-CP-11081 8228ZK11081	IT-CP-10081 8228ZK10081	IT-CP-10381 8228ZK10381	11.4	IT-ELB-10170 1111ZK10170	IT-ELB-20170 1111ZK20170
11.4	11.3	IT-CP-11082 8228ZK11082	IT-CP-10082 8228ZK10082	IT-CP-10382 8228ZK10382	11.5	IT-ELB-10171 1111ZK10171	IT-ELB-20171 1111ZK20171
11.5	11.4	IT-CP-11083 8228ZK11083	IT-CP-10083 8228ZK10083	IT-CP-10383 8228ZK10383	11.6	IT-ELB-10172 1111ZK10172	IT-ELB-20172 1111ZK20172
11.6	11.5	IT-CP-11084 8228ZK11084	IT-CP-10084 8228ZK10084	IT-CP-10384 8228ZK10384	11.7	IT-ELB-10173 1111ZK10173	IT-ELB-20173 1111ZK20173
11.7	11.6	IT-CP-11085 8228ZK11085	IT-CP-10085 8228ZK10085	IT-CP-10385 8228ZK10385	11.8	IT-ELB-10174 1111ZK10174	IT-ELB-20174 1111ZK20174
11.8	11.7	IT-CP-11086 8228ZK11086	IT-CP-10086 8228ZK10086	IT-CP-10386 8228ZK10386	11.9	IT-ELB-10175 1111ZK10175	IT-ELB-20175 1111ZK20175
11.9	11.8	IT-CP-11087 8228ZK11087	IT-CP-10087 8228ZK10087	IT-CP-10387 8228ZK10387	12	IT-ELB-10176 1111ZK10176	IT-ELB-20176 1111ZK20176
12	11.9	IT-CP-11088 8228ZK11088	IT-CP-10088 8228ZK10088	IT-CP-10388 8228ZK10388	12.1	IT-ELB-10177 1111ZK10177	IT-ELB-20177 1111ZK20177
12.1	12	IT-CP-11089 8228ZK11089	IT-CP-10089 8228ZK10089	IT-CP-10389 8228ZK10389	12.2	IT-ELB-10178 1111ZK10178	IT-ELB-20178 1111ZK20178

GROUP 05 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M10.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	11.1 – 13.1 mm
PIN ØD	11 – 13 mm
ELECTRODE ØD	11.2 – 13.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 16 mm	32.3 mm	40 mm / 40 mm	Uncooled M20x1.5 / 30 Water-cooled M26x1.5 / 36

IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.

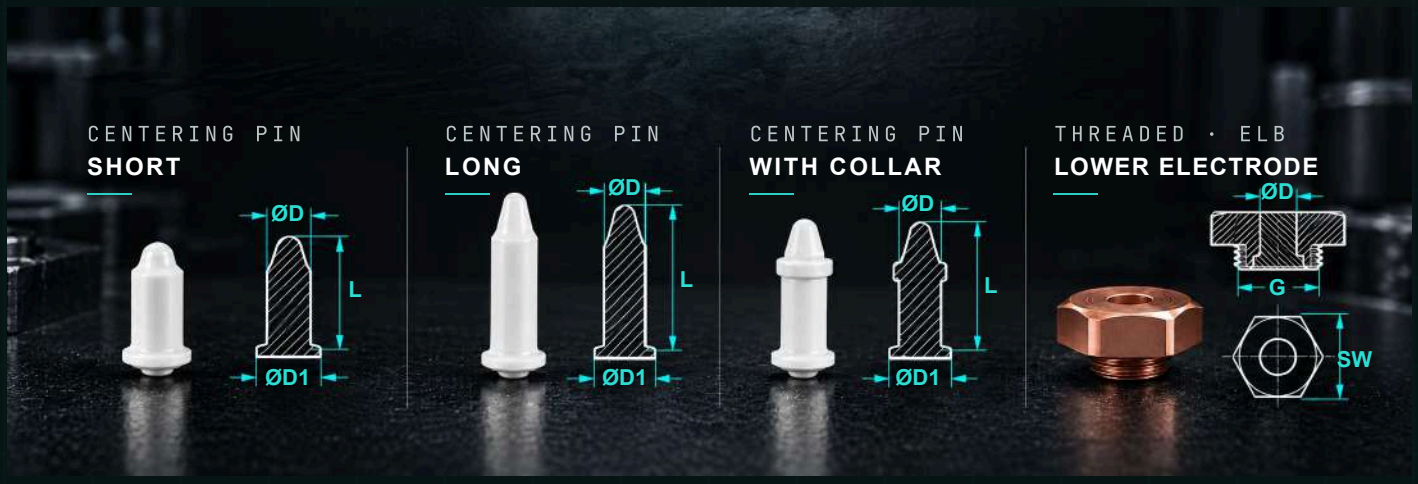
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 32.3 mm	PIN LONG L 40 mm	PIN WITH COLLAR L 40 mm	ELEC-TRODE ØD	ELB UNCOOLED G M20x1.5 · SW 30	ELB WATER-COOLED G M26x1.5 · SW 36
12.2	**12.1**	**IT-CP-11090** 8228ZK11090	**IT-CP-10090** 8228ZK10090	**IT-CP-10390** 8228ZK10390	**12.3**	**IT-ELB-10179** 1111ZK10179	**IT-ELB-20179** 1111ZK20179
12.3	**12.2**	**IT-CP-11091** 8228ZK11091	**IT-CP-10091** 8228ZK10091	**IT-CP-10391** 8228ZK10391	**12.4**	**IT-ELB-10180** 1111ZK10180	**IT-ELB-20180** 1111ZK20180
12.4	**12.3**	**IT-CP-11092** 8228ZK11092	**IT-CP-10092** 8228ZK10092	**IT-CP-10392** 8228ZK10392	**12.5**	**IT-ELB-10181** 1111ZK10181	**IT-ELB-20181** 1111ZK20181
12.5	**12.4**	**IT-CP-11093** 8228ZK11093	**IT-CP-10093** 8228ZK10093	**IT-CP-10393** 8228ZK10393	**12.6**	**IT-ELB-10182** 1111ZK10182	**IT-ELB-20182** 1111ZK20182
12.6	**12.5**	**IT-CP-11094** 8228ZK11094	**IT-CP-10094** 8228ZK10094	**IT-CP-10394** 8228ZK10394	**12.7**	**IT-ELB-10183** 1111ZK10183	**IT-ELB-20183** 1111ZK20183
12.7	**12.6**	**IT-CP-11095** 8228ZK11095	**IT-CP-10095** 8228ZK10095	**IT-CP-10395** 8228ZK10395	**12.8**	**IT-ELB-10184** 1111ZK10184	**IT-ELB-20184** 1111ZK20184
12.8	**12.7**	**IT-CP-11096** 8228ZK11096	**IT-CP-10096** 8228ZK10096	**IT-CP-10396** 8228ZK10396	**12.9**	**IT-ELB-10185** 1111ZK10185	**IT-ELB-20185** 1111ZK20185
12.9	**12.8**	**IT-CP-11097** 8228ZK11097	**IT-CP-10097** 8228ZK10097	**IT-CP-10397** 8228ZK10397	**13**	**IT-ELB-10186** 1111ZK10186	**IT-ELB-20186** 1111ZK20186
13	**12.9**	**IT-CP-11098** 8228ZK11098	**IT-CP-10098** 8228ZK10098	**IT-CP-10398** 8228ZK10398	**13.1**	**IT-ELB-10187** 1111ZK10187	**IT-ELB-20187** 1111ZK20187
13.1	**13**	**IT-CP-11099** 8228ZK11099	**IT-CP-10099** 8228ZK10099	**IT-CP-10399** 8228ZK10399	**13.2**	**IT-ELB-10188** 1111ZK10188	**IT-ELB-20188** 1111ZK20188

GROUP 06 · COORDINATED PIN/ELECTRODE SELECTION · 1/2

Ceramic pins + lower electrodes · M12.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	13.1 – 15.1 mm
PIN ØD	13 – 15 mm
ELECTRODE ØD	13.2 – 15.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 18 mm	34.1 mm	43 mm / 43 mm	Uncooled M22x1.5 / 30 Water-cooled M26x1.5 / 36

IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.

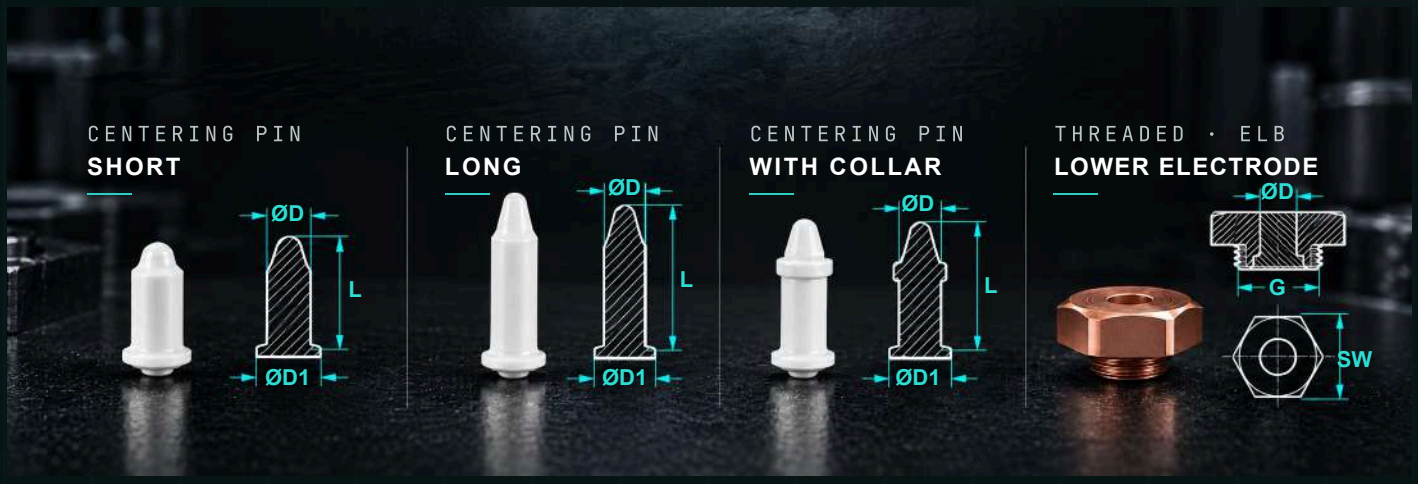
RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 34.1 mm	PIN LONG L 43 mm	PIN WITH COLLAR L 43 mm	ELEC-TRODE ØD	ELB UNCOOLED G M22x1.5 · SW 30	ELB WATER-COOLED G M26x1.5 · SW 36
13.1	13	IT-CP-11100 8228ZK11100	IT-CP-10100 8228ZK10100	IT-CP-10400 8228ZK10400	13.2	IT-ELB-10194 1111ZK10194	IT-ELB-20194 1111ZK20194
13.2	13.1	IT-CP-11101 8228ZK11101	IT-CP-10101 8228ZK10101	IT-CP-10401 8228ZK10401	13.3	IT-ELB-10195 1111ZK10195	IT-ELB-20195 1111ZK20195
13.3	13.2	IT-CP-11102 8228ZK11102	IT-CP-10102 8228ZK10102	IT-CP-10402 8228ZK10402	13.4	IT-ELB-10196 1111ZK10196	IT-ELB-20196 1111ZK20196
13.4	13.3	IT-CP-11103 8228ZK11103	IT-CP-10103 8228ZK10103	IT-CP-10403 8228ZK10403	13.5	IT-ELB-10197 1111ZK10197	IT-ELB-20197 1111ZK20197
13.5	13.4	IT-CP-11104 8228ZK11104	IT-CP-10104 8228ZK10104	IT-CP-10404 8228ZK10404	13.6	IT-ELB-10198 1111ZK10198	IT-ELB-20198 1111ZK20198
13.6	13.5	IT-CP-11105 8228ZK11105	IT-CP-10105 8228ZK10105	IT-CP-10405 8228ZK10405	13.7	IT-ELB-10199 1111ZK10199	IT-ELB-20199 1111ZK20199
13.7	13.6	IT-CP-11106 8228ZK11106	IT-CP-10106 8228ZK10106	IT-CP-10406 8228ZK10406	13.8	IT-ELB-10200 1111ZK10200	IT-ELB-20200 1111ZK20200
13.8	13.7	IT-CP-11107 8228ZK11107	IT-CP-10107 8228ZK10107	IT-CP-10407 8228ZK10407	13.9	IT-ELB-10201 1111ZK10201	IT-ELB-20201 1111ZK20201
13.9	13.8	IT-CP-11108 8228ZK11108	IT-CP-10108 8228ZK10108	IT-CP-10408 8228ZK10408	14	IT-ELB-10202 1111ZK10202	IT-ELB-20202 1111ZK20202
14	13.9	IT-CP-11109 8228ZK11109	IT-CP-10109 8228ZK10109	IT-CP-10409 8228ZK10409	14.1	IT-ELB-10203 1111ZK10203	IT-ELB-20203 1111ZK20203
14.1	14	IT-CP-11110 8228ZK11110	IT-CP-10110 8228ZK10110	IT-CP-10410 8228ZK10410	14.2	IT-ELB-10204 1111ZK10204	IT-ELB-20204 1111ZK20204

GROUP 06 · COORDINATED PIN/ELECTRODE SELECTION · 2/2

Ceramic pins + lower electrodes · M12.

Coordinated pin/electrode selection for resistance projection welding

HOLE Ø IN SHEET	13.1 – 15.1 mm
PIN ØD	13 – 15 mm
ELECTRODE ØD	13.2 – 15.2 mm
TECH. DOCS	IT-CP / IT-ELB



FIXED PIN DIMENSIONS	L · SHORT	L · LONG / WITH COLLAR	ELB · G / SW
ØD1 18 mm	34.1 mm	43 mm / 43 mm	Uncooled M22x1.5 / 30 Water-cooled M26x1.5 / 36

IT-… DGSI / IndustryTools part no. 8228… / 1111… Cross-reference no.

RECOMMENDED HOLE Ø IN SHEET	PIN ØD	PIN SHORT L 34.1 mm	PIN LONG L 43 mm	PIN WITH COLLAR L 43 mm	ELEC-TRODE ØD	ELB UNCOOLED G M22x1.5 · SW 30	ELB WATER-COOLED G M26x1.5 · SW 36
14.2	14.1	IT-CP-11111 8228ZK11111	IT-CP-10111 8228ZK10111	IT-CP-10411 8228ZK10411	14.3	IT-ELB-10205 1111ZK10205	IT-ELB-20205 1111ZK20205
14.3	14.2	IT-CP-11112 8228ZK11112	IT-CP-10112 8228ZK10112	IT-CP-10412 8228ZK10412	14.4	IT-ELB-10206 1111ZK10206	IT-ELB-20206 1111ZK20206
14.4	14.3	IT-CP-11113 8228ZK11113	IT-CP-10113 8228ZK10113	IT-CP-10413 8228ZK10413	14.5	IT-ELB-10207 1111ZK10207	IT-ELB-20207 1111ZK20207
14.5	14.4	IT-CP-11114 8228ZK11114	IT-CP-10114 8228ZK10114	IT-CP-10414 8228ZK10414	14.6	IT-ELB-10208 1111ZK10208	IT-ELB-20208 1111ZK20208
14.6	14.5	IT-CP-11115 8228ZK11115	IT-CP-10115 8228ZK10115	IT-CP-10415 8228ZK10415	14.7	IT-ELB-10209 1111ZK10209	IT-ELB-20209 1111ZK20209
14.7	14.6	IT-CP-11116 8228ZK11116	IT-CP-10116 8228ZK10116	IT-CP-10416 8228ZK10416	14.8	IT-ELB-10210 1111ZK10210	IT-ELB-20210 1111ZK20210
14.8	14.7	IT-CP-11117 8228ZK11117	IT-CP-10117 8228ZK10117	IT-CP-10417 8228ZK10417	14.9	IT-ELB-10211 1111ZK10211	IT-ELB-20211 1111ZK20211
14.9	14.8	IT-CP-11118 8228ZK11118	IT-CP-10118 8228ZK10118	IT-CP-10418 8228ZK10418	15	IT-ELB-10212 1111ZK10212	IT-ELB-20212 1111ZK20212
15	14.9	IT-CP-11119 8228ZK11119	IT-CP-10119 8228ZK10119	IT-CP-10419 8228ZK10419	15.1	IT-ELB-10213 1111ZK10213	IT-ELB-20213 1111ZK20213
15.1	15	IT-CP-11120 8228ZK11120	IT-CP-10120 8228ZK10120	IT-CP-10420 8228ZK10420	15.2	IT-ELB-10214 1111ZK10214	IT-ELB-20214 1111ZK20214